Learning Analytics Engagement and Outcomes Across Course Runs

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2026-01-13

`{r setup, include=FALSE} knitr::opts_chunk$set(echo = FALSE, message = FALSE, warning = FALSE) knitr::opts_knit$set(root.dir = normalizePath(".."))`

`{r load_project, include=FALSE} library(ProjectTemplate) load.project()`

1. Introduction:

Learning analytics examines learner behaviour data to understand participation, and outcomes in online courses. Understanding these patterns can help course teams analyse the course performance and identify areas where learner engagment may be improved.

This report analyses data from seven runs of **FutureLearn course** using the CRISP-DM framework. The analysis is of two cycles, the first cycle examines the overall learner outcomes across course runs and the second cycle analyses step-level engagement in order to identify where learners disengage during the course.

The report addresses the following research questions:

1. **How do learner outcomes(completion, unenrolment and certificate purchase)differ across course runs?**
2. **At which stages of the course do learners disengage, and are these patterns consistent across runs?**

2.CRISP-DM Cycle 1: Learner Outcomes Across Course Runs

2.1 BUsiness Understanding:

From a course delivery perspective, learner outcomes provide an overview of course effectiveness. Three outcomes are of interest to the stakeholders: if learners complete the course, number of unenrolment before completion, and if lerners purchase a certificate. Differences in these outcomes across course runs may indicate variation in learner engagement or course delivery conditions.

In addition, examining certificate purchase among learners who complete the course provides insight into post-completion behaviour and conversion to paid certification.

2.2 Data Understanding:

Cycle 1 uses learner level enrolment data from seven course runs. The dataset contains one record per learner per run, timestamps indicating course completion, unenrolment, and certificate purchase. Where timestamps are missing, the particular action is assumed not to have occured. These missing values therefore reflect learner behaviour rather than data quality issues.In total, the combined dataset includes several thousand learner records across all runs.

This analysis uses learner-level 'enrolments' data to identify completion, unenrolment, and certificate purchase outcomes across course runs. 'Step activity' data are included to examine how engagement changes across individual course steps and to identify points where leaner drop-off. Question response data are included to support learner-level analysis, while other datasets such as video statistics, archetype-survey responses, team members, leaving survey responses, sentiment data are excluded as they do not provide direct evidence of how learners progress through the course and disengagemnet across runs.

"'{r Table1, fig.cap = "Overview of datasets used in the anlysis", fig.align='center', echo = FALSE, fig.width = 6, fig.height = 3}

data_overview

```
 Table 1 provides an overview of the datasets used in the analysis including the number of rows and colu

## 2.3 Data Preparation:

Enrolment data from all the runs were combined into a single dataset, with the run number derived from t

Completion, certificate purchase, and unenrolment were treated as binary outcomes, and summary statstics

## 2.4 Modelling and Results

'''{r Figure1, fig.cap = "Learner outcomes by course run", fig.align='center', echo = FALSE, fig.width =
ggplot(plot_data, aes(x= factor(run), y= percent, fill= outcomes))+
  geom_col(position="dodge")+
  labs(x="course run",
       y= "percentage of learners (%)",
       title= "learner outcomes by course run",
       fill= "outcomes")
```

Figure 1 shows the percentage of learners who completed the course, unenrolled, or purchased a certificate across the seven runs. Completion rates vary substantially between runs, ranging from approximately 0.5% to 12.5%. **Run 1 records the highest completion rate at around 12.5%** while few other runs show completion rates below 2%

Unenrolment rates are consistently higher than completion rates. In some runs, unenrolment reaches **approximately 18-19%** indicating that a large proportion of learners disengage early. Certificate purchase rates remain low across all runs, generally below 1.5% of enrolled learners.

```
                  Table 2: Learner outcome summary by course run
```

`{r} plot_data`

Table 2 reports the numerical values underlying Figure 1 and confirms that learner outcomes vary across runs. To examine the relationship between completion and certificate purchase, purchase rates were recalculated among lerners who completed the course.

```
                Figure 2: Certificate purchase rate among completers by course runs
```

"'{r} ggplot(completers_by_run, aes(x= factor(run), y= purchase_rate_among_completers)) + geom_col(fill= "steelblue")+ labs(x= "Course run",665 y= "certificate purchase rate among completors (%)", title= "Certificate purchase rate among course completers by run")

Figure 2 shows that certificate purchase among completers varies across runs, ranging from below 5% to a

2.5 Evaluation

Cycle 1 shows that learner outcomes differ across course runs. Completion rates vary by more than 10 per

3.CRISP-DM Cycle 2 : Step-level Engagement and Disengagement

3.1 Business Understanding:

While Cycle 1 identifies variation in learner outcomes, it does not indicate where learners disengage du

3.2 Data Understanding:

Cycle 2 uses step-level activity data recording learner visits and completions for each step. Each recor

3.3 Data Preparation:

Step activity data from all runs were combined into a single dataset and ordered according to the course

3.4 Modelling and Results

Figure 3: Completion rate across the runs

```
```{r}
 #step level summary
step_summary <- step_activity2 %>%
  group_by(run, week_number, step_number) %>%
  summarise(visits= sum(visited),
            completions= sum(completed),
            completion_rate= completions/visits,
            .groups ="drop")
step_summary

#finding steps with low completion rate
step_summary %>%
  filter(visits >0 )%>%
  arrange(completion_rate) %>%
  head(10)

dim(step_summary)
head(step_summary)
summary(step_summary$completion_rate)
```

Figure 3 shows step completion rates across the course. Completion rates decline as learners progress through the steps, with several later steps showing completion rates below 20%.

Table 3 reports the number of vists, completions, and completion rates for each step, supporting the pattern observed in Figure 3.Figure 4: Steps with lowest completion rates across runs

Figure 4: Steps with Lowest completion rates

`{r} ggplot(low_completion_steps,        aes(x= factor(step_number),y= completion_pct))+ geom_col(fill = "orange")+   facet_wrap(~run)+   labs( x= "Step number",         y= "completion rate (%)",         title= "Lowest Completion steps by couse runs")` Figure 4 compares the steps with lowest completion rates across runs. Although the specific step numbers vary slightly between runs, low completion consistently occurs in similar sections of the course.
```

```
Table 4: Summary of Lowest completion rates
```

"'{r} low_completion_steps <- step_summary %>% filter(visits > 100) %>% arrange (completion_pct) %>% head(10)

low_completion_steps <- low_completion_steps %>% arrange(step_number)

low_completion_steps %>% select(run, week_number, step_number, completion_pct)

```
Table 4 lists the lowest-completion steps by run, allowing direct comparison of problamatic stages acros
```

```
              Figure 5: Retention curve for Each run
```
```{r}
ggplot(retention_curve, aes(x= max_step, y=active_pct, colour= factor(run)))+
  annotate("rect", xmin= 19.5, xmax = 21.5, ymin= -Inf, ymax= Inf, alpha =0.08)+
  geom_line(linewidth = 1, alpha= 0.9)+
  scale_y_continuous(limits= c(0,100), breaks= seq(0,100,10))+
  labs( x= "step number",
        y= "learners still active(%)",
        colour = "course run",
        title= " Learner retention (drop-off) curve by course run")+
  theme_minimal(base_size = 12)+
  theme(panel.grid.minor = element_blank())
```

Figure 5 shows the rentention curves for each run. In all runs, retention declines graduallly rather than showing a single sharp drop. By later steps, fewer than 20% of learners remain active, and in some runs this falls below 10%

```
        Figure 6: Stage where largest single-step drop-off occur
```

`{r} ggplot(largest_drop_off_by_run, aes(x=factor(run), y= dropoff_pct)) +   geom_point(size = 3) +   geom_text(aes(label = step_number), vjust =-0.8, size=3)+   labs(x= "Course run",   y= "largest single-step drop(percentage points)",   title= "Biggest drop-off point by course run",   subtitle= "Number above each point= step where the biggest drop occurs")+   theme_minimal()` Figure 6 shows the largest single step drop-off for each run. The largest drop-offs range from approximately 5 to 12 percentage points and occurs in the mid to later stages of the course.

"'{r}

"' Table 5 reports the step number and magnitude of the largest drop-off for each run.

3.5 Evaluation:

Cycle 2 indicates that learner disengagement increases as learners move through the course. Completion rates decline across steps, retention decreases steadily, and the largest drop-offs occur after initial engagemnet. These patterns are consistent across runs, suggesting that disengagemnet is more closely related to course progression than to run specific factors.

Final Recommendations and Conclusions:

The analysis shows clear differences in learner outcomes and engagement across course runs, with completion and retention declining consistently as learners progress through the course. Step-lebvel patterns indicate taht disengagement builds gradually than being driven by a single factor, suggesting structural or workload-related challenges in later stages.

Taargeted redesign of low-completion steps, improved pacing of the course,and clear communication of course benefits, may help improve retention and conversion among active learners. Overall,the combined run-levl and step-level analysis, provides insights into where course design changes are likely to help strengthen learner engagements and outcomes.